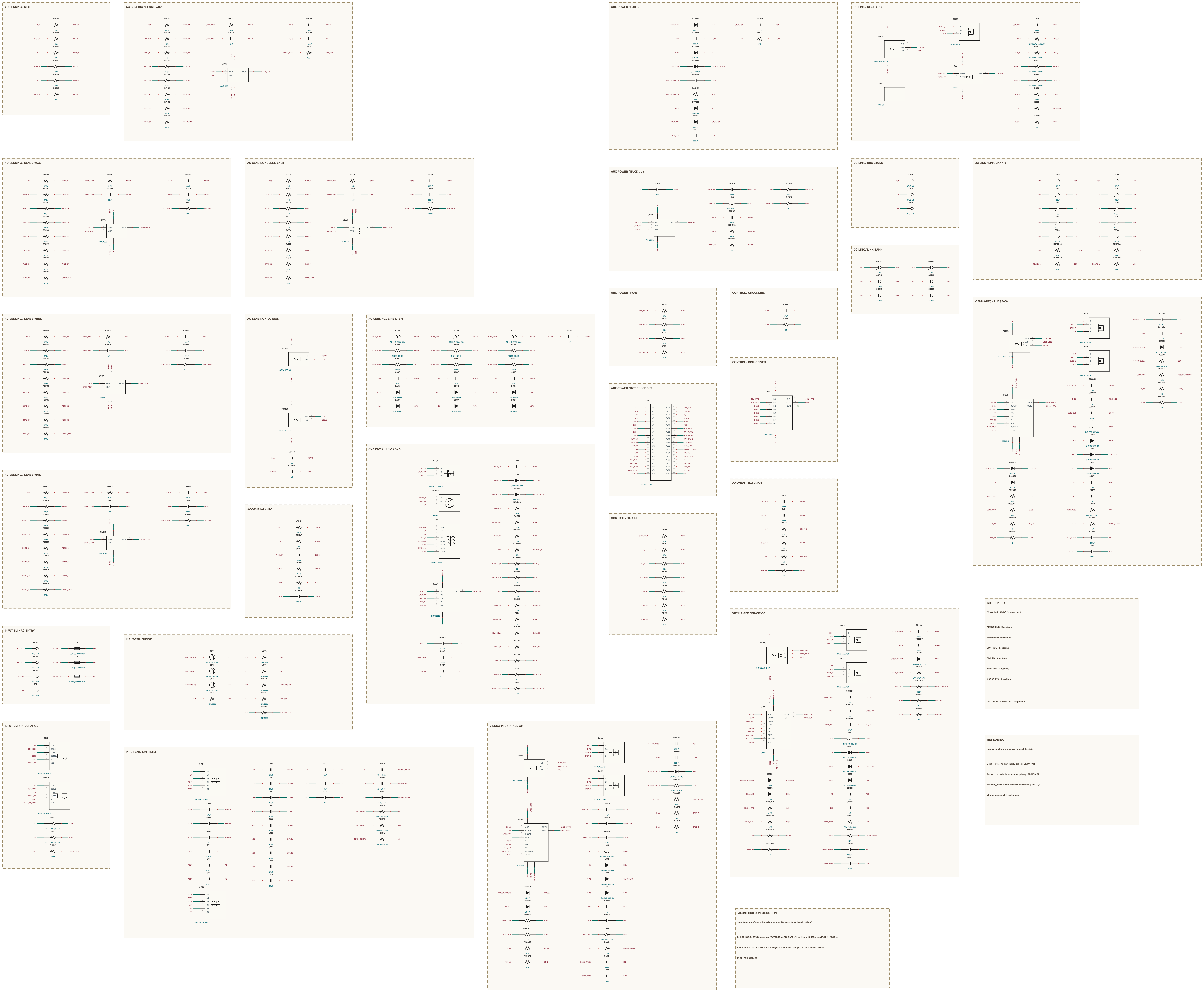

SHEET INDEX

50 kW liquid AC-DC (lower) - 1 of 3

AC-SENSING - 9 sections

AUX-POWER - 5 sections

CONTROL - 4 sections

DC-LINK - 4 sections

INPUT-EMI - 4 sections

VIENNA-PFC - 3 sections

rev D.4 - 29 sections - 342 components

NET NAMING

internal junctions are named for what they join

U<ref>_<PIN> node at that IC pin e.g. UIVOA_VINP

R<stem>_M midpoint of a series pair e.g. RBALTA_M

R<stem>_<nm> tap between R<stem>n/m e.g. RV1D_01

all others are explicit design nets

MAGNETICS CONSTRUCTION

identity per docs/magnetics.md (turns, gap, litz, acceptance lines live there)

D1 LA0-LC0: 5x T79 26u sendust (CATALOG AL37), N=24 +/-1 lot trim -> L0 107uH, >=45uH @129.5A pk

EMI: CMC1 + 12x X2 4.7uF in 3 star stages + CMC2 + RC damper; no AC-side DM chokes

Cr at TANK sections

DC-Modules 50 kW liquid — AC-DC board (Vienna PFC)

DC-Modules 50 kW liquid - board AC-DC (lower), sheet 1 of 3

2026-09-06 · rev D.4